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!pip install opencv-python-headless scikit-image matplotlib numpy

import os
import warnings
import numpy as np
import matplotlib.pyplot as plt

import cv2

from skimage import io, feature, img_as_ubyte

warnings.filterwarnings("ignore", category=UserWarning)

def show_two_images(img1, title1, img2, title2, cmap1='gray',
cmap2='gray', figsize=(12,6)):
    plt.figure(figsize=figsize)
    plt.subplot(1,2,1)
    plt.title(title1)
    plt.axis('off')
    if img1.ndim == 2:
        plt.imshow(img1, cmap=cmap1)
    else:
        plt.imshow(cv2.cvtColor(img1, cv2.COLOR_BGR2RGB))
    plt.subplot(1,2,2)
    plt.title(title2)
    plt.axis('off')
    plt.imshow(img2, cmap=cmap2)
    plt.show

IMAGE_PATH = "quazi.jpg"

if not os.path.exists(IMAGE_PATH):
    raise FileNotFoundError(f"Image not found at '{IMAGE_PATH}'. Put
your image in the notebook folder or chnage the IMAGE_PATH.")

img_color = cv2.imread(IMAGE_PATH)
img_gray = cv2.imread(IMAGE_PATH, cv2.IMREAD_GRAYSCALE)

if img_color is None or img_gray is None:
    raise ValueError("Failed to load image.")

plt.figure(figsize=(6,6))
plt.title("Original (RGB)")
plt.axis('off')
plt.imshow(cv2.cvtColor(img_color, cv2.COLOR_BGR2RGB))
plt.show()

low_thresh = 100
high_thresh = 200

edges = cv2.Canny(img_gray, low_thresh, high_thresh)

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show_two_images(img_gray, "Original (grayscale)", edges, f"Canny edges
(thr={low_thresh},{high_thresh})")

image_for_lbp = img_as_ubyte(io.imread(IMAGE_PATH, as_gray=True))

P = 8
R = 1
method = 'uniform'

lbp = feature.local_binary_pattern(image_for_lbp, P=P, R=R,
method=method)
show_two_images(image_for_lbp, "Original (for LBP)", lbp, f"LBP(P={P},
R={R}, method='{method}')")

# SIFT may require opencv-contrib-python; use fallback to ORB if SIFT
not present.
gray = cv2.cvtColor(img_color, cv2.COLOR_BGR2GRAY)

keypoints = None
image_with_keypoints = None

try:
    # Try SIFT
    sift = cv2.SIFT_create()
    keypoints = sift.detect(gray, None)
    image_with_keypoints = cv2.drawKeypoints(img_color, keypoints,
None, flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS)
    print(f"Detected {len(keypoints)} SIFT keypoints.")
except Exception as e:
    print("SIFT not available or failed; using ORB instead.
(Exception: {}).format(e))
    orb = cv2.ORB_create(nfeatures=500)
    keypoints = orb.detect(gray, None)
    image_with_keypoints = cv2.drawKeypoints(img_color, keypoints,
None, flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS)
    print(f"Detected {len(keypoints)} ORB keypoints.")

plt.figure(figsize=(8,6))
plt.title("Keypoints (SIFT or ORB)")
plt.axis('off')
plt.imshow(cv2.cvtColor(image_with_keypoints, cv2.COLOR_BGR2RGB))
plt.show()

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